

FACULTY OF ENGINEERING

COMP342/EE329

Computer Networks

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NETWORK DESIGN PROJECT

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1. INTRODUCTION

Computer network refers to a set of computers that share resources which are being provided by network nodes. To communicate with each other, computers make use of common communication protocols over digital interconnection. To make a network connection between the devices, the use of a cable or wireless media is very necessary. When the interconnections are made, to allow data to transfer between the devices with the use of the network, there are various protocols that can be used. These protocols include TCP/IP. Within a network design plan, the devices and the network should be configured very well in order to attain the defined requirements. Problems may arise, and they need solutions before the issue becomes huge. A good network design plan must be able to provide the solutions. This project focuses on the network design for European University of Lefke's Faculty of Engineering. The focus is to create a secure network infrastructure that accommodates administration staff offices, laboratories, seminar rooms and lecture theatre in the multiple floors with a total of ninety-nine users. The project will involve sub netting a Class C network (204.15.5.0) to meet the host requirements using Cisco Packet Tracer. A star topology is going to be used to build this network. A star topology is there is a central router which will be making sure that there is a well organised data flow and centralized control. Each floor will be connected to this central router. The router is one of the key components of the network. It acts as the gateway because it will be connecting subnets on each floor. To allow proper internal communication, two switches in each floor are linked with a copper straight through cables. CLI commands are used to configure switches and routers during the implementation. The implementation was done carefully to make sure everything is correctly set up and also to make sure proper IP allocation is done to each subnet. To confirm that the devices can communicate through the network flawlessly, a simple ping test is done. PCs and laptops with static IP addresses connect to switches are the hosts. In the organization of the network, sub netting plays a major role. To create smaller and more manageable subnets that match the needs of each floor, borrowing bits from the host portion of the IP addresses is done. The design is built with real users in mind. It supports everyday needs like fast, reliable internet for students and staff, smooth video conferencing and secure access to important academic and administrative data. High speed Ethernet ensures low lag which is crucial for activities like online learning. Security is also a top priority with firewalls and encryption protecting sensitive information such as student records and research data.

2. NETWORK DESIGN

NETWORK TOPOLOGY

INTRODUCTION TO NETWORK TOPOLOGY

A physical or logical connection of the devices of a computer network is referred to as a network topology. It focuses on the connection between computers, cables, switches, and other equipment including the direction of data communication among them. The main purpose of a network topology is to administer network structure for proper and flowless communication and management. There are various topologies, for example, star, bus, ring, and mesh. These topologies support different needs based on cost, reliability, and performance. Topology is understood to make it easier to maintain, scale, and troubleshoot networks

IMPORTANCE IN COMMUNICATION AND DATA FLOW

Network topology is of very high importance since it shows how devices are connected.

It helps to design how data will move from one device to another.

A good topology reduces delay and speeds up the transmission of data.

It helps in simple resolution and repair of network problems.

Certain topologies, like mesh, offer better reliability and avoid single points of failure.

It also helps in minimizing costs by choosing the best cabling and device configuration.

With proper topology, the network is easy to expand without many changes.

Overall, it improves how well the network works and keeps communication in good shape.

TYPE OF NETWORK TOPOLOGY USED

Network topology is the way in which computers and devices are linked to a network. It shows where the data transfers between devices. These include the star, bus, ring, and mesh. Each type has its own speed, cost, and reliability. However, in this project we are going to be using star topology because of its main advantage the failure of one device does not affect the rest of the network, enhancing overall reliability. Centralized management simplifies the process of identifying and resolving issues. New devices can be added without reconfiguring the entire network.

Star Topology

- All devices are connected to one central switch or hub in star topology.

- The hub is the main controller which controls the flow of data.
- Each device has a direct connection to the hub, hence easy to follow faults.
- It's reliable—if one of the devices fails, others are not affected.
- But if the central hub fails, the entire network goes down.
- It is widely used in schools, offices, and home networks due to its ease of use and speed
- Star topology is a topology where all devices (computers or printers, for instance) are plugged into a main hub or switch. The main hub acts as the controller and deals with the distribution of data from one device to another. Whenever a device needs to forward information to another device, it passes it through to the hub first, and it forwards the data to the respective destination.

BLOCK DIAGRAM.

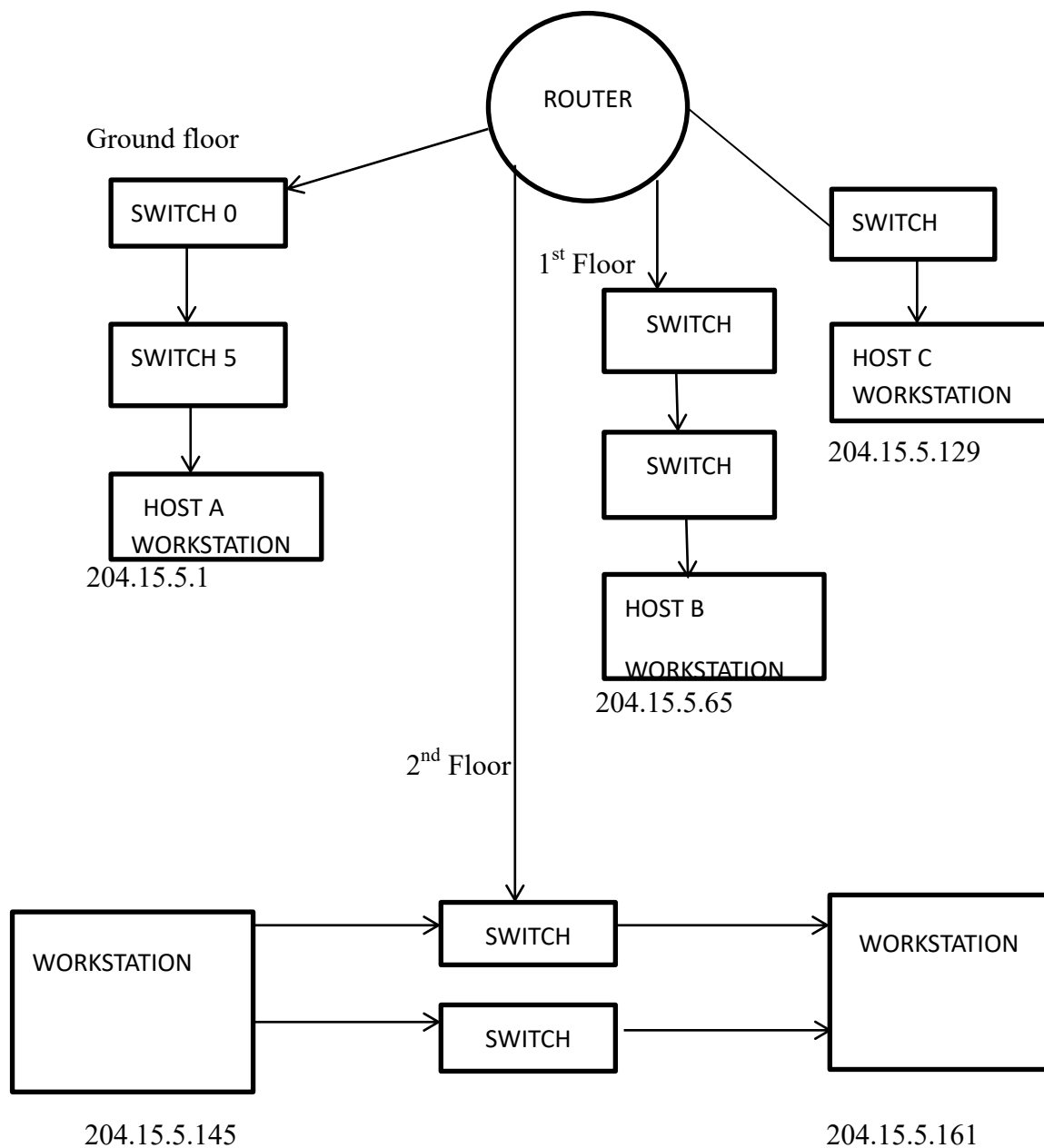


Figure 1 Block diagram for the network design

System block diagram for the star topology shows a centralized device, usually a switch or a router, connected to various end devices like PCs. The PCs communicate through the central router, not directly with each other. Arrows among the PCs face towards and away from the router, showing data flow. A transmitter sends data to the router and the router forwards it to the target PC or to a receiver. This setup offers structured communication and network management at ease.

FLOW CHART FOR THE NETWORK DESIGN

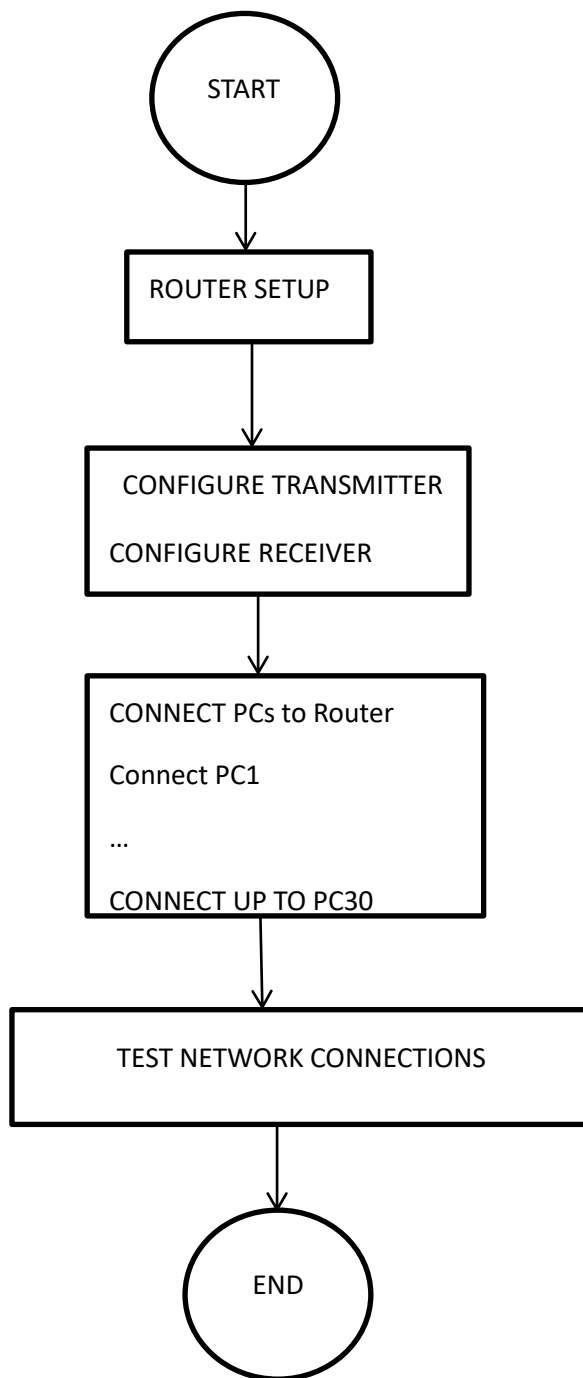


Figure 2 Flowchart for the network design

LAYOUT DESIGN AND TESTING TROUBLESHOOTING COMMON ISSUES

This process of establishing the network is started by planning the topology accordingly, choosing the right topology and placing devices appropriately. After physical connections have been made through cables or wireless means, individual devices are then configured

with settings like IP addresses and security configurations. After setup, the network is verified for connectivity between devices by using commands like ping. Performance is then validated by speed tests and monitoring tools to ensure data exchange is smooth.

ADVANTAGES AND DISADVANTAGES OF THE STAR NETWORK TOPOLOGY

Advantages

Improved performance and reduced data collisions. Within a star topology, packets of data travel in a linear path directly from the source to the hub, and then directly to the destination device. With the direct flow of communication, the chances of data collisions like in other topologies of bus or ring are reduced. Being centralized, the hub helps control traffic in an efficient way, resulting in better overall performance, especially in handling multiple data transmissions simultaneously. In our project, continuous and high-speed communication is required, and star topology offers the reliability and performance to accomplish that.

Disadvantages

This point of failure introduces a vulnerability which can impact the overall reliability of the project. For instance, if our project is so designed to receive real-time data transmission or emergency communications, even a temporary disruption due to hub failure may lead to data loss, delays, or system downtime. Besides, the cost and technical complexity of ensuring the reliability of the hub center—such as the use of backup systems or redundant channels can add to the total cost and technical complexity of the project.

3. USER REQUIREMENTS

1. Interactivity

Interactivity refers to the ability of a network to make room for a real-time and seamless communication between users and services. In an environment like academic environment ,for example in this project European University of Lefke engineering department, interactivity is very important, because it makes sure that

teaching, learning, administrative work, and research activities takes place without any disturbances. There is need for ability to access online course materials, as well as learning platforms for example, Moodle and video conferencing tools for example Microsoft Teams. On the same note, faculty members like the lecturers need access to grading systems, internal communication platforms, emails, cloud storage, and other shared applications. For this to be done efficiently, there is a need of a network that supports low latency and fast data transfer rates.

The network must be built with modern high-speed Ethernet. This makes room for fast file transfers, as well as a flawless multimedia streaming, and responsive remote access for the network to meet interactivity needs. For data to flow smoothly between users, high quality switches and routers are very necessary. It also reduces congestion. The network should not slow down and should be able to accommodate a number of concurrent users, for example when a lecturer uploads lecture notes to a shared drive.

Interactivity should also be seen in internal tools for example, library management systems, student record databases, and email servers. There should never be crash or slow network on the systems because they should be accessible at all times with fast response. This is mostly crucial during course registration times, exams periods and assignment submissions times when many students and lectures (users) are active at the same time.

In overall, the network must be fast and very interactive to make modern academic environment easy and interesting. Every user, including the students and the staff in the administration office must be able to access the resources they need without any glitches or inconvenience in terms of network. High interactive network allows user satisfaction.

2. Reliability

Reliability is another important feature of any computer network, especially when it is being used by many users. Reliability means that the network should function effectively, continuously with zero to no interruptions or glitches. This is very crucial for important activities such as administrative processing, research simulations, online teaching, and student collaboration in the case of a university.

A reliable network is built with good planned network topology. It is also built with the use of durable, equipment, Cisco routers and switches. The network should be very reliable to the extent that a failure in one part of the network does not bring the whole network down. For a

network to be reliable, the use of proper cabling, making sure there are secure power supplies, and the maintenance of backup hardware is very important.

In addition, to ensure network reliability, proper configuration is very important. To avoid switching loops, implementing features like spanning tree protocol (STP) can be very useful. Redundancy protocols like HSRP (Hot Standby Router Protocol) or VRRP (Virtual Router Redundancy Protocol) can be very useful in making sure that if one router fails, another router takes over automatically. Important services and servers should be backed up. To protect the key hardware from things like failures of power, UPS (Uninterruptible Power Supply) systems should be used.

To keep track of the system's health in real time, the use of network monitoring tools is very essential. To help ensure fast response to issues, the use of tools like SNMP (Simple Network Management Protocol) or network management software can be very helpful because they alert the administrator of any failures.

The objective of network reliability is to ensure an uninterrupted service to users. The network will be able to meet user demands reliably and without failure throughout the use of solid design principles, redundant systems, and proper monitoring.

4. Adaptability

Adaptability refers to the ability of the network to handle changes over time. For example, adding new users, upgrading equipment and supporting new applications. A flexible network is very important in an academic environment where technology and user needs are changing constantly over time.

For example, an engineering building may only be using 100 computers connected in the current time, but in the future it might need 50 or 100 more devices due to maybe new labs. So the network should have the ability to accommodate such changes without a redesign. To

ensure this, the use of switches with extra ports, planning subnets with more IP address space and using routers that can handle higher throughput is very essential and beneficial.

Another side of adaptability is being able to integrate new technologies. For example, the network must be able to handle new data types and traffic patterns, if the university starts to use cloud computing tools, Internet of Things (IoT) devices and artificial intelligence services in classrooms or labs to separate departments or labs logically, yet still using the same physical infrastructure, VLANs (Virtual LANs) can be used. This makes it easier to manage and upgrade.

Software and services are also included in adaptability. Regular updates to security protocols must be supported by the network. Access permissions and software tools must also be supported by the network. The network should also support VPN (Virtual Private Network) due to the rising demand for remote access. The network should be able to secure remote desktop services, and cloud storage access.

To add more, to decrease the costs and disruption during upgrades, the use of modular designs, where various parts of the network can be changed or added/expanded without affecting the rest. For example, upgrading a core switch should not need reconfiguring every lab switch or IP range.

Conclusively, adaptability makes sure that as user needs change over time, the network remains useful and relevant. The university avoids costly overhauls in future, if it designs a scalable and flexible network today.

5. Security

Network security is important in protecting sensitive and important data, ensuring the privacy of users, and disabling unauthorized access. In a university environment, network security is especially important because the network accommodates academic records, exam materials, research data, and administrative files. Lack of network security usually results in data loss, theft of identity, or damage to the institution's reputation.

Ensuring physical security is the very first important thing to do. Avoiding access to networking equipment for example, like routers and switches to unauthorized personnel and only allowing authorised personnel. Logical security measures must then be implemented. Strong passwords must be put on router and switches. Also management interfaces should be secured by the use of SSH (Secure Shell) instead of Telnet. To filter traffic coming from the Internet, the use of firewalls is very important. Access Control Lists (ACLs) can be implemented on routers to reduce traffic between internal departments. For example, only Admin PCs can access or open certain financial servers. To separate traffic logically, VLANs can also be used. For instance, having student labs on separate VLAN from staff offices.

Topping up more to access control, another important thing is encryption. Sensitive information for instance exam papers or grades should not be travelling in plain text across the network. To ensure that data is private and secure, such technologies like SSL/TLS for web applications and VPNs for remote connections can be used.

Additionally, regular updates and patch management are equally of great importance. Old firmware or software can have high risks of hackers exploiting. Network administrators should use IDS/IPS (Intrusion Detection/Prevention Systems) to stay updated with patches and monitor for threats.

To add more, user training and policies are also part of network security. Using weak passwords or even clicking a phishing link can affect even the best technical system. To help strengthen the overall security, conducting awareness programs and enforcing strong password policies can be very helpful.

In conclusion, securing the network is about protecting people and their information not just protecting machines. The foundation for trust in the digital learning environment is a wellsecured network.

4. APPLICATION REQUIREMENTS

Student Management Application Systems

i. Student Information System (SIS)

A Student Information System (SIS) manages or works with student data management. It has responsibilities such as enrolling students, and personal information management. The SIS platform enables access to data. There are things like academic progress tracking, as well as class scheduling, course management. Self service portals allows the students to have access to their records and have the ability independently to manage their academic journey. There is confidentiality of the student information due to security and data management ii. Course Registration

For academic planning, efficient course registration and scheduling are very crucial. This allows students to enroll in courses, manage and access their timetables. These systems also enables administrators, perform resource allocation, making sure there is optimal utilization of classrooms. iii. Grading

To assess the performance of a student,grading systems are very necessary. They make room for instructors to put student grades, calculate their GPAs, and make their transcripts. iv.

Student Portal

Student portals, with a personalized interface, allows students to have full access to their academic information including course materials, and grades. It also allows them to communicate with the faculty.

Administration Management Application Systems

i. Human Resources Management

They manage the personal information of employees, including payroll, and performance assessment. They also manage the recruitment of employees. There self service portals which enables staff to manage personal information and access their payslips. ii. Financial

Management Systems

These systems are mainly targetted on budgeting, accounting, and financial reporting. Working with SIS and HRM makes sure there is correct financial information related to tuition fees, salaries, and operational costs.

5. Facilities Management

This system enforces the scheduling of physical resources, for example classrooms and laboratories. Combining with scheduling systems facilitates optimal allocation of classrooms and laboratories.

iv. Communication and Collaboration Tools

These tools enable communication among staff, faculty, and students. They enable the sharing of information through emails, messaging.

Applications Running on Student Machines and Laboratories

i. Office Productivity Suites

There are very important applications that allow the creating of documents, data analysis, spread sheets and many more academic tasks. These applications include Microsoft Office and Google Workspace. ii. Programming and Development Tools

For computer and software engineering courses, Integrated Development Environments (IDEs) and compilers for languages are very important for example Java, and C++. iii. Simulation and Modeling Software

For students to model and simulate some engineering systems, they need tools like MATLAB and Simulink especially for electrical engineering.

iv. Computer - Aided Design (CAD) Software

CAD applications like AutoCAD or SolidWorks allows students to make more correct drawings and 3D models in engineering and other faculties hence they are equally important.

v. Virtual Laboratory system

Virtual laboratories are very useful especially in remote learning areas. They give simulated environments for experiments which is very flexible. **vi.** Learning Management Systems

For online learning, platforms like Moodle helps in course content, assignments and assessments. This is very flexible for both learners and instructors. The use of antivirus systems ensures the safety of academic data because security of student machines is very crucial and critical hence unauthorized programs must be prevented.

IMPLEMENTATION

IP sub netting is a process used in networking to divide a larger network into smaller manageable segments called subnets.

IP Address

A unique identifier assigned to each device on an network consisting of the two parts the network portion and host portion. IP consists of 32 bits.

Subnet Mask

A number that specifies which part of IP address is the network and which one is the host.

Sub netting

The act of applying a subnet mask to an IP address and create smaller sub networks within a large network .It is important to do sub netting because it reduces broadcast traffic, isolates network segments for security and makes IP address allocation more efficient.

In our case to design a network class C of 204.15.5.0 we have to subnet the network. In class C network we have a standard subnet mask of 255.255.255.0. It has 256 address which is from 0- 255.

In IP sub netting we borrow bits which means taking bits from host portion. The first bits are network bits and last 8 bits are for the host address. However on every network address, the first and last IP address are not used, that is we should reserve 204.15.5.0 to be for the network ID and 204.15.5..255 to be the host.

Subnets are assigned in descending order of size in this case all subnets for the floors, 30 host are allocated first. The remaining address space is divided into subnets for smaller floors. Subnets have a systematic approach where each subnet starts right after the previous one.

Subnets calculation

/27subnet

$$2^{(32-27)} - 2 = 32 - 2 \\ = 30$$

/28 subnet

$$2^{(32-28)} - 2 = 16 - 2 \\ = 14$$

SUBNET TABLE

Floor	Needed Host	Subnet Address	Subnet Mask	IP address Range	Broadcast Range
Ground floor: Staff Offices	30	204.15.5.0	255.255.255.224	204.15.5.1-204.15.5.30	204.15.5.31
Ground floor labs and lecturer room	Reserved	204.15.5.32	255.255.255.224	204.15.5.33-204.15.5.62	204.15.5.63
1st floor: Laboratories	30	204.15.5.64	255.255.255.224	204.15.5.65-204.15.5.94	204.15.5.95
2nd floor: Seminar Room and lecturer Theatre	25	204.15.5.96	255.255.255.224	204.15.5.97-204.15.5.126	204.15.5.127

1st floor: Staff Offices	14	204.15.5.128	255.255.255.240	204.15.5.129- 204.15.5.142	204.15.5.143
2nd floor: staff offices	Small	204.15.5.144	255.255.255.240	204.15.5.145 - 204.15.5.158	204.15.5.159
2nd floor: Micro labs, seminar and lecture rooms	Small	204.15.5.160	225.255.255.240	204.15.5.161 - 204.15.5.174	204.15.5.175

Table 1 Showing subnet mask and IP address range

CISCO NETWORK DESIGN AND IMPLEMENTATION

The design and configuration of a University LAN network with 3 floors using Cisco devices for a Class C network 204.15.5.0. The network has multiple subnets for different floors and departments with different host requirements and efficient IP addressing.

Cisco Network Design Components

- Star topology per floor: Central Control
- Router: It as gateway or connecting point of a network to any networks, in this case it is a routing device between subnets on each floor that forwards data packets between computer networks
- Switches: Two switches per floor connected by copper through cables for internal communication.
- PCs and Laptops: Connected to switches on respective floors with static IP addresses
- CABLES: For connections

EQUIPMENT AND SOFTWARE

- CISCO PACKET TRACER
- 1 ROUTER
- 7 SWITCHES
- 72PCs AND 28 LAPTOPS (hosts)
- Copper straight through cables

ROUTER PORTS	CONNECTED FLOORS	IPs	SUBNET MASK
Ground floor	GigabitEthernet0/0	204.15.5.1	255.255.255.224
1st Floor	GigabitEthernet0/1	204.15.5.65	255.255.255.224
1st Floor	FastEthernet0/0/0	204.15.5.129	255.255.255.240
2nd Floor	GigabitEthernet0/2	204.15.5.145	255.255.255.240

Table 2 Showing Router Interface allocations

Switch interface allocations

Floor	Switch Name	Workgroup	Connections	Switch PORTS
Ground floor	Switch 0 and Switch 3	Admin, staff offices, lecture theatre/rooms and seminar	Router gig0/0 Switch3	Fa0/1-Router Fa0/2-Switch3 Fa0/3-0/24 PCs

1st Floor	Switch 2 and Switch 4	Laboratories (research, Projects)	Router gig0/1 Switch4	Fa0/1-Router Fa0/2-Switch4 Fa0/3-0/24 PCS
1st Floor	Switch 6	Staff offices	Router Fa0/0/0	Fa0/1-Router Fa0/2-0/24 Laptops
2nd Floor	Switch1 and Switch5	Microlabs, Staff offices and seminar	Router gig0/2 Switch5	Fa0/1-Router Fa0/2-Switch5 Fa0/3-0/24 PCs & Laptops

Table 3 Switch interface allocations

- Switch ports used for inter-switch connections are using Fa0/2 and Fa0/1
- Router interfaces connect directly to the main switch on each floor.
- The Switch6 on the 1st floor is a separate subnet for staff offices requiring a different subnet mask.

Network Operation and Data Communication (University LAN Design)

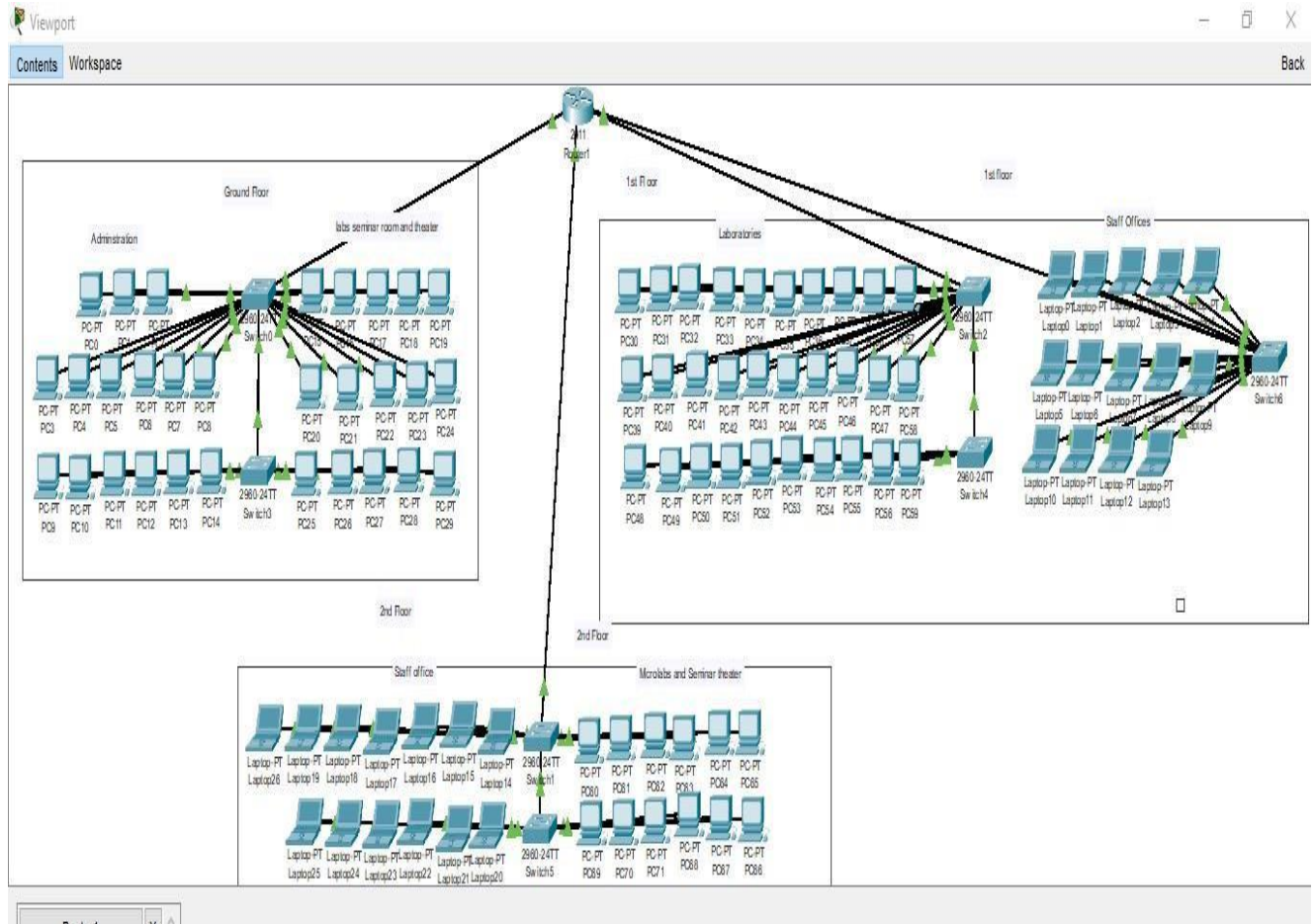
This network is designed as a LAN based hierarchical Star Topology with three floors each connected through switches to a central router.

Devices in the same Subnet communicate directly through their switch. Then devices in different subnets will communicate through the router which forwards packets between different subnets. The default gateways in this case are determined by the router interface IP of its subnet. The hosts in this network use static IP addresses for better control.

To sum up, this network contains Intra-Subnet communication for all PCs on the same floor, whereby the packets never go to router since devices are on the same subnet. Secondly,

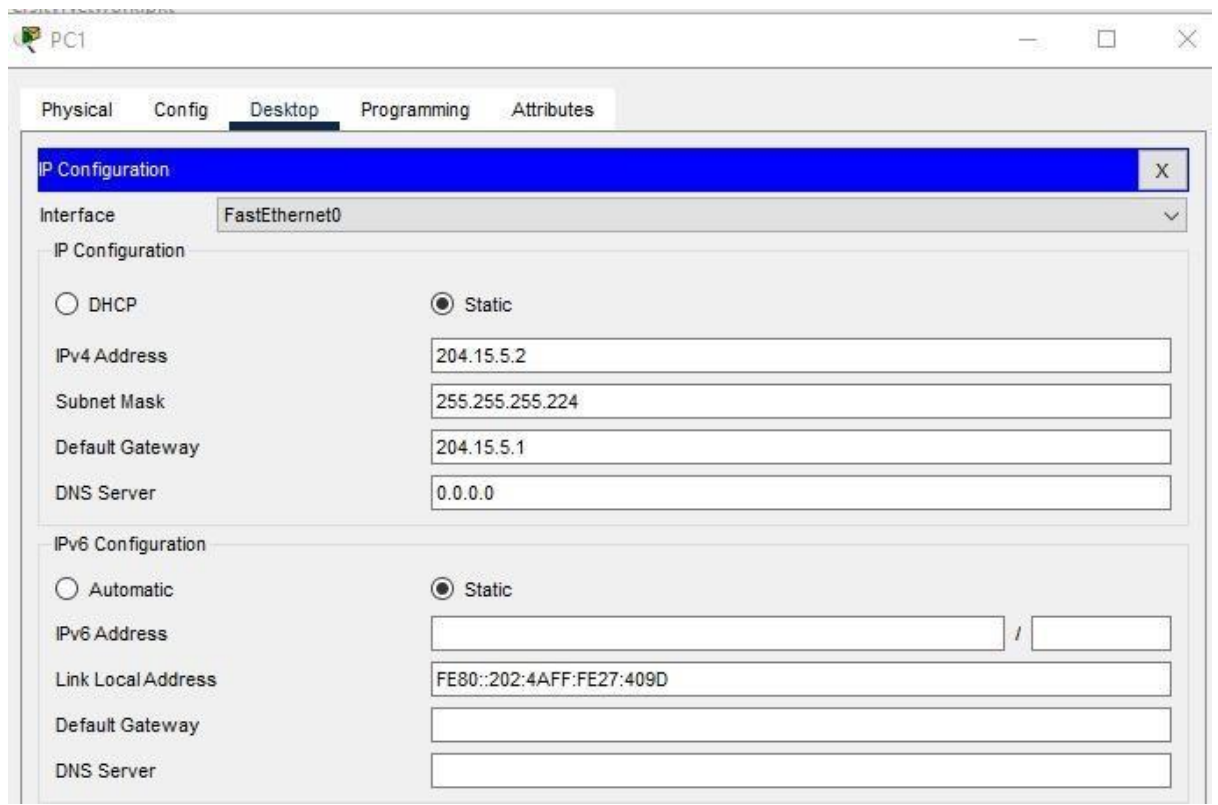
InterSubnet Communication which is where they are different floors, the router will act as a medium to forward packets. In this network all communication is bidirectional

NETWORK DESIGN



- Each rectangle represents a floor with its hosts connected to switches and the switch connected to the router
- Ground floor (2 switches, 30 hosts (PCs)) and First Floor (3 switches, 1 Switch uplink with 30 hosts (PCs) and last remaining switch with 14 hosts (Laptops) and lastly Second Floor with 25 hosts (12 PCs and 13 Laptops) connected to 2 switches (Uplink)

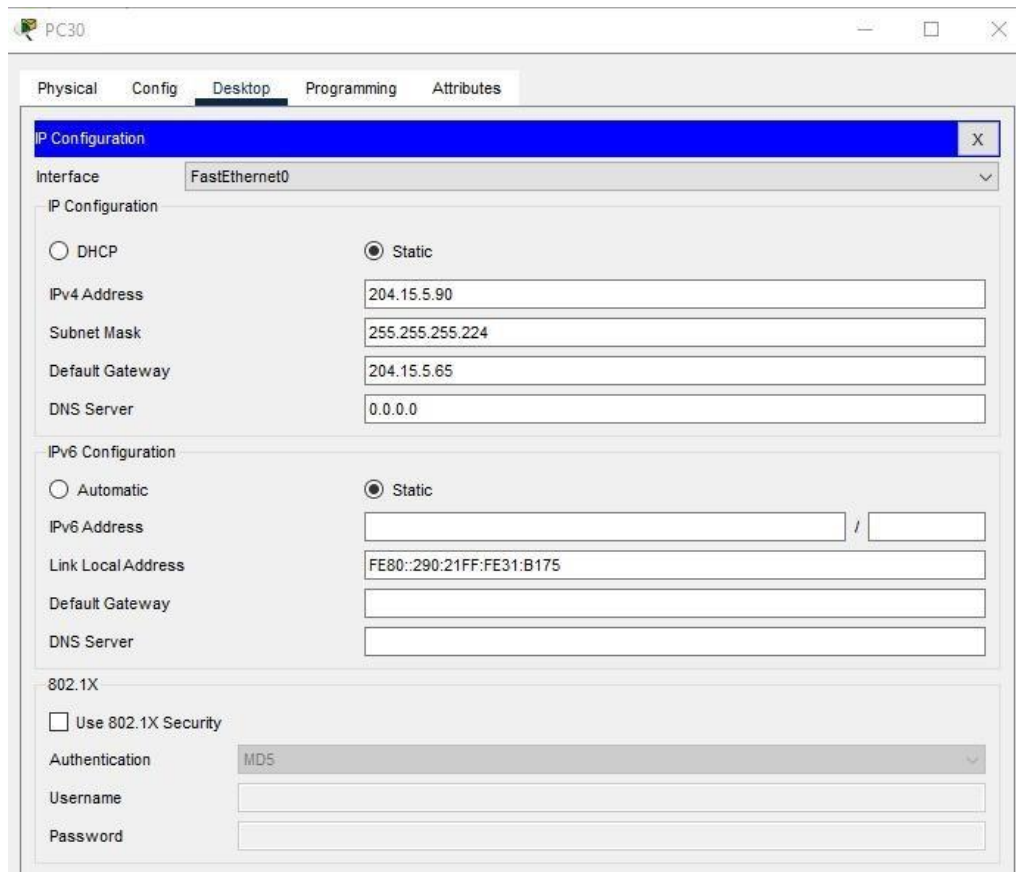
IP ADDRESS ALLOCATIONS



The screenshot shows the 'IP Configuration' window for PC1. The 'Desktop' tab is selected. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 fields are filled with: IPv4 Address: 204.15.5.2, Subnet Mask: 255.255.255.224, Default Gateway: 204.15.5.1, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The IPv6 fields are: IPv6 Address (empty), Link Local Address: FE80::202:4AFF:FE27:409D, Default Gateway (empty), and DNS Server (empty).

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	204.15.5.2
Subnet Mask	255.255.255.224
Default Gateway	204.15.5.1
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	
☒ Static	
IPv6 Address	
Link Local Address	FE80::202:4AFF:FE27:409D
Default Gateway	
DNS Server	

Figure 3 IP ADDRESS FOR THE GROUND FLOOR



The screenshot shows the 'IP Configuration' window for PC30. The 'Desktop' tab is selected. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 fields are filled with: IPv4 Address: 204.15.5.90, Subnet Mask: 255.255.255.224, Default Gateway: 204.15.5.65, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is selected. The IPv6 fields are: IPv6 Address (empty), Link Local Address: FE80::290:21FF:FE31:B175, Default Gateway (empty), and DNS Server (empty). At the bottom, under '802.1X', the 'Use 802.1X Security' checkbox is unchecked. The 'Authentication' dropdown is set to 'MD5'. The 'Username' and 'Password' fields are empty.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	
☒ Static	
IPv4 Address	204.15.5.90
Subnet Mask	255.255.255.224
Default Gateway	204.15.5.65
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	
☒ Static	
IPv6 Address	
Link Local Address	FE80::290:21FF:FE31:B175
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

Figure 4 IP ADDRESS FOR PC30 IN FIRST FLOOR

The screenshot shows the 'IP Configuration' window for PC60. The 'Desktop' tab is selected. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 fields are filled with: IPv4 Address: 204.15.5.161, Subnet Mask: 255.255.255.240, Default Gateway: 204.15.5.161, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The IPv6 fields are: IPv6 Address (empty), Link Local Address: FE80::260:5CFF:FE9C:24, Default Gateway (empty), and DNS Server (empty).

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	204.15.5.161
Subnet Mask	255.255.255.240
Default Gateway	204.15.5.161
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::260:5CFF:FE9C:24
Default Gateway	
DNS Server	

Figure 5 IP ADDRESS FOR PC60

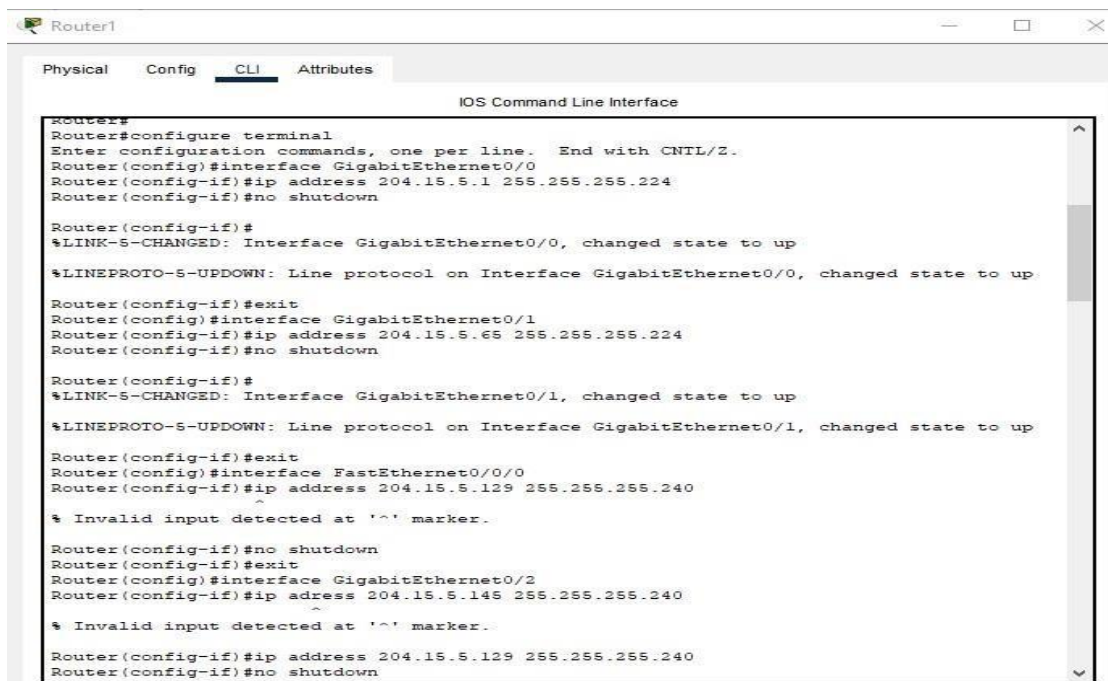
This screenshot is identical to Figure 4, showing the 'IP Configuration' window for PC60. The 'Desktop' tab is selected. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 fields are filled with: IPv4 Address: 204.15.5.161, Subnet Mask: 255.255.255.240, Default Gateway: 204.15.5.161, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The IPv6 fields are: IPv6 Address (empty), Link Local Address: FE80::260:5CFF:FE9C:24, Default Gateway (empty), and DNS Server (empty).

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	204.15.5.161
Subnet Mask	255.255.255.240
Default Gateway	204.15.5.161
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::260:5CFF:FE9C:24
Default Gateway	
DNS Server	

Figure 6 IP ADDRESS FOR PC60 IN THE SECOND FLOOR

ROUTER CONFIGURATION

- This is the process of setting up and managing a router to ensure it can effectively direct data between different computer networks thus configuring the router with IPs for each subnet this may be necessary to allow inter-subnet communication between different floors thus Configuration allows data to flow properly between devices and across the internet or other networks. It helps routers to know how to manage traffic.
- In this case, the configuration was done in the Command Line Interface of the router, this process involves entering a series of commands to define the router behaviour for instance
- Enabling the router
- Configuring interfaces
- Assigning IP addresses
- Setting up routing protocols



The screenshot shows a window titled 'Router1' with tabs for 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The terminal output shows the following commands and responses:

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 204.15.5.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 204.15.5.65 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#interface FastEthernet0/0/0
Router(config-if)#ip address 204.15.5.129 255.255.255.240
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip address 204.15.5.145 255.255.255.240
Router(config-if)#no shutdown
Router(config-if)#ip address 204.15.5.129 255.255.255.240
Router(config-if)#no shutdown
```

Figure 7 Router Configuration done in the project

```

Router#write memory
Building configuration...
[OK]
Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	204.15.5.1	YES	manual	up	up
GigabitEthernet0/1	204.15.5.65	YES	manual	up	up
GigabitEthernet0/2	204.15.5.129	YES	manual	up	up
FastEthernet0/0/0	unassigned	YES	unset	up	up
FastEthernet0/0/1	unassigned	YES	unset	up	down
FastEthernet0/0/2	unassigned	YES	unset	up	down
FastEthernet0/0/3	unassigned	YES	unset	up	down
GigabitEthernet0/1/0	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```

Router#

```

The routers CLI Command also shows you the status of all interfaces that is network ports whether they're up, have an IP address, or are shut down

SWITCH CONFIGURATIONS

A switch-to-switch link was established through copper straight through cables (Fast Ethernet). In this case each switch needed to be connected therefore it was done through the following for all:

The screenshot shows a network switch configuration window with two panes. The left pane displays the CLI commands entered, and the right pane shows the resulting output.

```

Switch0
Physical Config CLI Attributes
IOS Command Line Interface

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered
with Switch FastEthernet0/1 (1).

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Switch0
Switch0(config)#interface
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered
with Switch FastEthernet0/1 (1).

% Incomplete command.
Switch0(config)#interface fa0/1
Switch0(config-if)#no shutdown
Switch0(config-if)#exit
Switch0(config)#interface fa0/2
Switch0(config-if)#no shutdown
Switch0(config-if)#exit
Switch0(config)#end
Switch0#
%SYS-5-CONFIG_I: Configured from console by console

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, c

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#

```

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Switch3
Switch3(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with
Switch0 FastEthernet0/2 (10).

Switch3(config)#interface fa0/1
Switch3(config-if)#no shutdown
Switch3(config-if)#exit
Switch3(config)#interface fa0/2
Switch3(config-if)#no shutdown
Switch3(config-if)#exit
Switch3(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with
Switch0 FastEthernet0/2 (10).
exit
Switch3(config)#end
Switch3#
%SYS-5-CONFIG_I: Configured from console by console

Switch3#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with
Switch0 FastEthernet0/2 (10).

```

Figure 8 Switch configurations

PING TEST RESULTS

To demonstrate successful connections between devices in the same subnets and different subnets
Ping results for Ground Floor (PC21) and first floor (Laptop0)

Command Prompt

```
Pinging 204.15.5.21 with 32 bytes of data:

Reply from 204.15.5.21: bytes=32 time<1ms TTL=128
Reply from 204.15.5.21: bytes=32 time<1ms TTL=128
Reply from 204.15.5.21: bytes=32 time<1ms TTL=128
Reply from 204.15.5.21: bytes=32 time<1ms TTL=128

Ping statistics for 204.15.5.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 204.15.5.17

Pinging 204.15.5.17 with 32 bytes of data:

Reply from 204.15.5.17: bytes=32 time<1ms TTL=128
Reply from 204.15.5.17: bytes=32 time<1ms TTL=128
Reply from 204.15.5.17: bytes=32 time<1ms TTL=128
Reply from 204.15.5.17: bytes=32 time<1ms TTL=128

Ping statistics for 204.15.5.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 204.15.5.5

Pinging 204.15.5.5 with 32 bytes of data:

Reply from 204.15.5.5: bytes=32 time=1ms TTL=128
Reply from 204.15.5.5: bytes=32 time<1ms TTL=128
Reply from 204.15.5.5: bytes=32 time=1ms TTL=128
Reply from 204.15.5.5: bytes=32 time=1ms TTL=128

Ping statistics for 204.15.5.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
```

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 204.15.5.129

Pinging 204.15.5.129 with 32 bytes of data:

Reply from 204.15.5.129: bytes=32 time<1ms TTL=128
Reply from 204.15.5.129: bytes=32 time=3ms TTL=128
Reply from 204.15.5.129: bytes=32 time<1ms TTL=128
Reply from 204.15.5.129: bytes=32 time<1ms TTL=128

Ping statistics for 204.15.5.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>ping 204.15.5.136

Pinging 204.15.5.136 with 32 bytes of data:

Reply from 204.15.5.136: bytes=32 time<1ms TTL=128
Reply from 204.15.5.136: bytes=32 time=34ms TTL=128
Reply from 204.15.5.136: bytes=32 time<1ms TTL=128
Reply from 204.15.5.136: bytes=32 time=1ms TTL=128

Ping statistics for 204.15.5.136:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 34ms, Average = 6ms

C:\>ping 204.15.5.140

Pinging 204.15.5.140 with 32 bytes of data:

Reply from 204.15.5.140: bytes=32 time=1ms TTL=128
Reply from 204.15.5.140: bytes=32 time<1ms TTL=128
Reply from 204.15.5.140: bytes=32 time<1ms TTL=128
Reply from 204.15.5.140: bytes=32 time<1ms TTL=128
```

PING RESULTS FOR SECOND FLOOR

```
Command Prompt

Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 204.15.5.159

Pinging 204.15.5.159 with 32 bytes of data:

Reply from 204.15.5.152: bytes=32 time=83ms TTL=128
Reply from 204.15.5.156: bytes=32 time=83ms TTL=128
Reply from 204.15.5.155: bytes=32 time=83ms TTL=128
Reply from 204.15.5.151: bytes=32 time=83ms TTL=128
Reply from 204.15.5.150: bytes=32 time=83ms TTL=128
Reply from 204.15.5.148: bytes=32 time=83ms TTL=128
Reply from 204.15.5.154: bytes=32 time=83ms TTL=128
Reply from 204.15.5.147: bytes=32 time=83ms TTL=128
Reply from 204.15.5.149: bytes=32 time=83ms TTL=128
Reply from 204.15.5.146: bytes=32 time=83ms TTL=128
Reply from 204.15.5.153: bytes=32 time=103ms TTL=128
Reply from 204.15.5.151: bytes=32 time=1ms TTL=128
Reply from 204.15.5.153: bytes=32 time=1ms TTL=128
Reply from 204.15.5.156: bytes=32 time=1ms TTL=128
Reply from 204.15.5.155: bytes=32 time=1ms TTL=128
Reply from 204.15.5.148: bytes=32 time=31ms TTL=128
Reply from 204.15.5.154: bytes=32 time=33ms TTL=128
Reply from 204.15.5.147: bytes=32 time=33ms TTL=128
Reply from 204.15.5.149: bytes=32 time=44ms TTL=128
Reply from 204.15.5.146: bytes=32 time=44ms TTL=128
Reply from 204.15.5.150: bytes=32 time=44ms TTL=128
Reply from 204.15.5.152: bytes=32 time=44ms TTL=128
Reply from 204.15.5.151: bytes=32 time<1ms TTL=128
Reply from 204.15.5.152: bytes=32 time=1ms TTL=128
Reply from 204.15.5.153: bytes=32 time=1ms TTL=128
```

```
PC61

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 204.15.5.161

Pinging 204.15.5.161 with 32 bytes of data:

Reply from 204.15.5.161: bytes=32 time<1ms TTL=128
Reply from 204.15.5.161: bytes=32 time<1ms TTL=128
Reply from 204.15.5.161: bytes=32 time<1ms TTL=128
Reply from 204.15.5.161: bytes=32 time<1ms TTL=128

Ping statistics for 204.15.5.161:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 204.15.5.169

Pinging 204.15.5.169 with 32 bytes of data:

Reply from 204.15.5.169: bytes=32 time<1ms TTL=128
Reply from 204.15.5.169: bytes=32 time=1ms TTL=128
Reply from 204.15.5.169: bytes=32 time<1ms TTL=128
Reply from 204.15.5.169: bytes=32 time<1ms TTL=128

Ping statistics for 204.15.5.169:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 204.15.5.170

Pinging 204.15.5.170 with 32 bytes of data:

Reply from 204.15.5.170: bytes=32 time<1ms TTL=128
Reply from 204.15.5.170: bytes=32 time<1ms TTL=128
Reply from 204.15.5.170: bytes=32 time=1ms TTL=128
Reply from 204.15.5.170: bytes=32 time<1ms TTL=128
```

FAILED PING

```
C:\>ping 204.25.5.70

Pinging 204.25.5.70 with 32 bytes of data:

Reply from 204.15.5.65: Destination host unreachable.
Reply from 204.15.5.65: Destination host unreachable.
Reply from 204.15.5.65: Destination host unreachable.
Reply from 204.15.5.65: Destination host unreachable.

Ping statistics for 204.25.5.70:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 204.15.5.93

Pinging 204.15.5.93 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 204.15.5.93:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

C:\>
```

CONCLUSION

This project successfully demonstrated and carefully implemented network that meets the needs of a modern university environment. By sub netting a class C IP address and using Cisco Packet Tracer to simulate the setup. A functional and efficient LAN is designed to connect three floors of European University Of Lefke's faculty of engineering building each with different host requirements and usage scenarios. The star topology with a central router that managed traffic between floors and switches handling local communication provided a clean and scalable structure. The configuration process was done through CLI which provides experience in assigning IP addresses, setting up interfaces and making the devices can communicate to each other in the same subnet and across different ones. The use of static IP addresses offered control over device management. Connectivity was tested using ping commands which showed that most devices were able to communicate as intended. The successful ping responses confirmed that subnets were correctly configured and routing between them worked. There was one ping that failed this is to show us that even well designed systems can have small issues making it important to do troubleshooting and thorough testing. Lastly, this project provided more than just a working network but also a clear understanding of how to use sub netting to design a network and also about device configurations.

RECOMMENDATIONS

The network infrastructure served the needs of connection, effectiveness, and organization. Nevertheless, there are some areas that can be improved to make it stronger, easier to manage, and prepared for future requirements. First of all, the one failed ping test shows that a small issue maybe a misconfigured interface .unplugged cable or disabled port can be overlooked during installation so we need to inspect the devices involved to make sure that everything is properly plugged and connected. Static IP addresses worked for this project, but they can be a chore to manage in environments like student labs where hardware is constantly changing. It would be time efficient and less prone to mistakes to use a DHCP server to assign IP addresses automatically. Security is always a concern at a university where sensitive data is kept and transmitted. The current network can be improved with more security measures, and the creation of VLANs would keep the traffic isolated among departments. This would make it challenging for intruders to access information they don't need to see.

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APPENDIX



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